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Broadband and compact polarization rotator-splitter based on adiabatic tapers and a 2×2 MMI coupler on the thin-film lithium niobate platform

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We present a broadband and compact polarization rotator-splitter (PRS) implemented on the thin-film lithium niobate (TFLN) platform, utilizing two adiabatic tapers, a phase shifter and a general-interference-based 2×2 multimode interference (MMI) coupler. The device employs adiabatic mode conversion and multimode interference mechanisms to achieve mode conversion and polarization splitting within a compact length of $228 \mu\text{m}$. Experimental results indicate that the fabricated device exhibits low insertion losses, with 0.2 dB for the TE_0 mode and 0.6 dB for the TM_0 mode at a wavelength of 1550 nm. Furthermore, the device demonstrates remarkable tolerance to fabrication variations within $\pm 4 \mu\text{m}$, highlighting its potential for practical implementations. This work paves the way for polarization-multiplexed transceivers, reconfigurable optical networks, and quantum photonic systems on thin-film lithium niobate platforms. © 2025 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

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The exponential growth of global data traffic, driven by the emergence of advanced technologies such as 5G networks, cloud computing, and artificial intelligence, has placed unprecedented demands on optical communication systems [1]. To address the escalating requirements for higher transmission capacity and enhanced energy efficiency, advanced multiplexing techniques, such as wavelength-division multiplexing (WDM) and polarization-division multiplexing (PDM), have emerged as essential techniques [2]. Among these techniques, PDM leverages the two orthogonal polarization states of light to effectively double the channel capacity without requiring additional spectral resources, thereby establishing itself as a fundamental component of modern coherent optical communication systems [3]. However, the practical implementation of PDM is highly dependent on compact, high-performance polarization-handling devices, such as polarization rotators, splitters, and beam combiners. These devices must operate with minimal insertion loss, broad bandwidth, and robust fabrication tolerance to ensure reliable and efficient signal transmission. The development of

such advanced optical polarization-handling components is crucial for advancing the capabilities of PDM and supporting the increasing demands of high-capacity optical communication networks.

Thin-film lithium niobate (TFLN) has recently gained significant attention, with high-performance devices rapidly advancing on this platform. This progress is primarily attributed to its outstanding optical properties, including high refractive index contrast and large electro-optic coefficients [4]. The TFLN platform has enabled the development of diverse high-performance photonic devices, including electro-optical modulators [5], wavelength converters [6], high-stability optical oscillators [7], and broadband optical frequency combs [8]. Furthermore, the heterogeneous integration of semiconductor light sources [9], semiconductor optical amplifiers [10], and photodetectors [11] enables the realization of multi-functional lithium niobate integrated chips.

Despite these advantages, the inherent birefringence of lithium niobate results in anisotropic characteristics. Light propagating along different crystallographic axes exhibits distinct refractive indices, thereby posing a challenge for the design of polarization-insensitive systems. Consequently, it is essential to develop on-chip polarization management components tailored for TFLN platforms [12], thereby maximizing their potential in next-generation optical systems. A polarization rotator-splitter (PRS) is a key component of polarization management devices. To date, PRSs on TFLN platforms have been successfully demonstrated [13]. However, these devices are typically large, with some reaching millimeters in length, which poses a challenge for high-density, multi-functional integrated circuits.

To address these challenges, we propose and demonstrate a novel PRS on the TFLN platform. The device utilizes adiabatic mode conversion and multimode interference principles to simultaneously achieve mode conversion and polarization splitting within a compact length of $228 \mu\text{m}$. Through numerical simulation and experimental verification, we designed a polarization rotator-splitter with two tapered waveguide structures, achieving low insertion loss (< 1.0 dB for the TE_0 and TM_0 modes) and high polarization extinction ratio (> 12.6 dB for both TE_0 and TM_0 modes) across the C- + L-band (1520–1630 nm). This work paves the way for polarization-multiplexed transceivers,

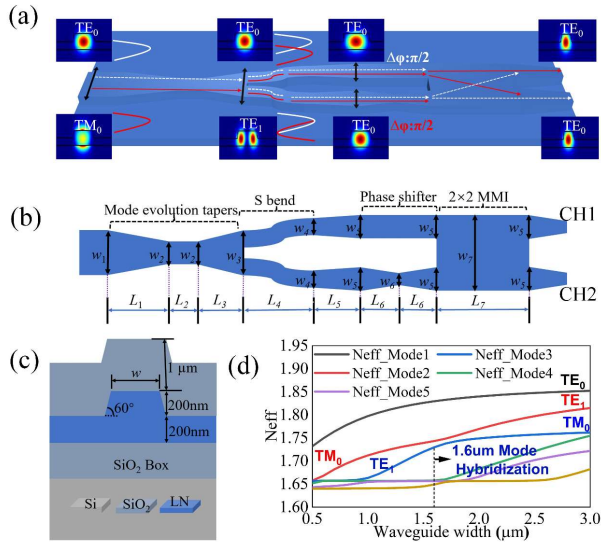


Fig. 1. (a) Schematic illustration of the PRS on TFLN and (b) top view of the PRS (not to scale). (c) Cross section and (d) N_{eff} for the lowest-order modes at a wavelength of 1550 nm with respect to the waveguide width.

reconfigurable optical networks, and quantum photonic systems on thin-film lithium niobate platforms.

As shown in Fig. 1(a), the TE_0 mode propagates along the tapered waveguide and is then split into two beams by a Y-splitter, producing two TE_0 modes with the same phase. For the TM_0 input, the TM_0 mode is efficiently converted to the TE_1 mode by the adiabatic taper and then split into two TE_0 modes with a π phase difference. For the TE_0 input, the TE_0 mode remains unchanged and is split into two TE_0 modes with no phase difference. In both cases, a phase difference of $\pi/2$ is introduced before entering the multimode interference (MMI). For TE_0 , the phase difference is $\pi/2$, resulting in TE_0 being output from CH2, minimizing cross talk in CH1. For TM_0 , the phase difference is $-\pi/2$, resulting in TM_0 being output from CH1, minimizing cross talk in CH2. As illustrated in Fig. 1(b), the proposed PRS consists of five components: two adiabatic tapers, a straight waveguide, an s bend, a phase shifter, and a 2×2 MMI coupler based on general interference. The PRS is designed on a 400 nm X-cut with the main propagation axis along the optical Y axis. As shown in Fig. 1(c), the device is formed by ridge waveguides etched to a depth of 200 nm and cladded with 1- μm SiO_2 . Considering the current fabrication process, the sidewall tilt angle is set to 60° .

To achieve the mode conversion from TM_0 to TE_1 , a half-etched ridge waveguide is used, which is asymmetric in the vertical direction due to the etching conditions, as shown in Fig. 1(c). As a result, mode hybridization occurs within a specific range of the waveguide width for TM_0 and TE_1 . Figure 1(d) shows the calculated effective refractive index of the five lowest-order eigenmodes at 1550 nm as a function of the ridge waveguide width. It can be observed that the effective indices of the TM_0 and TE_1 modes converge within the width range of approximately 1.5–2.1 μm , with an anti-crossing near the center (Fig. 1(d)). At the anti-crossing point, where the waveguide width is 1.6 μm , the difference in effective refractive indices between TE_1 and TM_0 is only 0.013. Considering that polarization purity increases rapidly outside the range

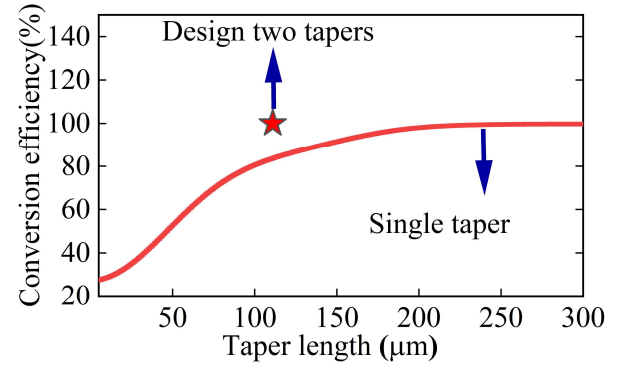


Fig. 2. Conversion efficiency of a single taper versus the taper length. The red mark indicates the conversion result of the proposed four-taper design.

of 1.5–2.1 μm , mode hybridization is significantly weakened. Therefore, the taper widths in the mode evolution section of our design are set to 1.5 μm and 2.1 μm .

A single linear taper requires a longer length; therefore, multiple tapers are employed for TM_0 – TE_1 conversion in this work. To verify the above theory, Lumerical MODE is employed to calculate the conversion efficiency of a single taper (with width varying from 1.5 to 2.1 μm) as a function of the taper length, as shown in Fig. 2. Considering that nonadiabatic evolution may occur if the taper is too short, we calculated the efficiency starting from a length of 5 μm . An efficiency of 99% requires a taper length of over 220 μm (Fig. 2). As shown in Fig. 2, the conversion process is divided into two parts: fast (<100 μm) and slow (>100 μm). When the taper length is 100 μm , the conversion efficiency exceeds 80%. This phenomenon suggests that multiple short tapers can be used to reduce the overall length requirement, as each taper transition can remain within the fast section. Furthermore, as seen from Eq. (3) (see Supplement 1, Section 1), mode conversion depends not only on the two tapered sections but also on the length of the straight waveguide. The conversion efficiency can be flexibly tuned by varying the lengths of multiple tapered and straight waveguides, enabling nearly 100% conversion while minimizing the overall device length.

Based on the above discussion, to achieve optimal mode conversion, the widths at both ends of the tapers are set to 2.1 μm ($w_1 = w_3$), the width of the straight waveguide is set to 1.5 μm (w_2), and a double-taper and a straight waveguide structure is adopted. The particle swarm optimization method is used to maximize conversion efficiency while minimizing the total length. As a result, we achieved a conversion efficiency of 99.8% at a wavelength of 1550 nm, with the lengths of the two tapers and the straight waveguide being 59 μm (L_1), 45 μm (L_3), and 7 μm (L_2), respectively, resulting in a total length of only 111 μm (red markers in Fig. 2). Simulation results show that for devices designed with the above parameters, the transmission rate for TE_0 input reaches as high as 99.9%.

As shown in Fig. 1(a), these waveguides divide the converted TE_1 mode into TE_0 modes with a phase difference of π in the upper and lower arms while also dividing the input TE_0 mode into TE_0 modes with no phase difference between the arms. Before entering the 2×2 MMI, an additional phase difference of $\pi/2$ is required. When the TE_0 mode is input, it is output from CH2 due to the $\pi/2$ phase difference. When the TM_0 mode is input, it is output from CH1 due to the $-\pi/2$ phase difference.

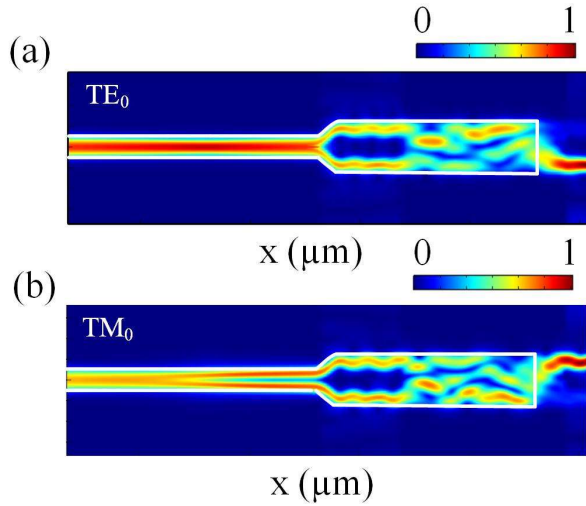


Fig. 3. Simulated electric field magnitude distribution of the PRS for (a) TE_0 and (b) TM_0 inputs.

This results in the realization of the mode rotation and splitting functions for both TE_0 and TM_0 . Consequently, the design of the phase shifter is crucial. The design of the phase shifter is shown in the phase shifter section of Fig. 1(b). The length of the phase shifter is set to $20\ \mu\text{m}$ ($2L_6$). At this length, the phase shift and transmission spectrum of the phase shifter vary with the waveguide width. When the upper arm waveguide width is set to $1.5\ \mu\text{m}$ (w_5) and the lower arm waveguide width is $0.8\ \mu\text{m}$ (w_6), the phase difference between the optical modes propagating in the upper and lower arms reaches $\pi/2$ (see Supplement 1 Section 2). The width and length of the 2×2 MMI coupler is $5\ \mu\text{m}$ (w_7) and $76\ \mu\text{m}$ (L_7).

Based on the above design parameters, the broadband performance of the PRS device is investigated. Figures 3(a) and 3(b) show the electric field magnitude distributions of the PRS for the TM_0 and TE_0 inputs at $1550\ \text{nm}$, respectively, clearly illustrating the polarization rotation and splitting functions achieved by the proposed structure.

Remarkably, Figs. 4(a) and 4(b) show the calculated transmission spectra of the TM_0 and TE_0 modes at the two output ports for the TE_0 and TM_0 mode inputs within a bandwidth of $1500\text{--}1630\ \text{nm}$. As demonstrated in Fig. 4(a), for the TE_0 mode input, the transmission of CH2 port remains nearly unchanged, with a loss of less than $0.7\ \text{dB}$. The polarization extinction ratio (PER) remains high (above $33\ \text{dB}$) across the entire bandwidth, as the TE_0 mode does not undergo mode hybridization within the device. As shown in Fig. 4(b), for the TM_0 mode input, the transmission of CH1 port decreases slightly, with a loss of less than $1.6\ \text{dB}$. Compared to the TE_0 input, the PER drops slightly in the $1500\text{--}1630\ \text{nm}$ range, but remains above $29\ \text{dB}$. This is due to the mode conversion of the TM_0 mode throughout the device, with the conversion efficiency at different wavelengths slightly affecting both the PER and the losses.

The length tolerance of the MMI at $1550\ \text{nm}$ is discussed, as shown in Fig. 5. As seen in Fig. 5, within a range of $\pm 4\ \mu\text{m}$, the insertion loss of the TE_0 mode remains below $0.7\ \text{dB}$, with the corresponding PER above $32\ \text{dB}$. The insertion loss of the TM_0 mode remains below $0.8\ \text{dB}$, with the corresponding PER above $27\ \text{dB}$. The PER of the TM_0 mode is obviously slightly lower than that of the TE_0 mode. This is because the TM_0 mode has a

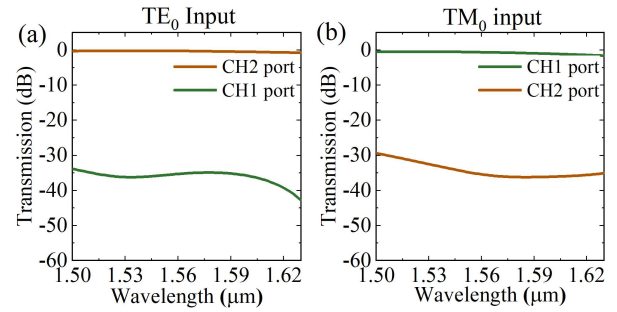


Fig. 4. Simulated transmission spectra at CH1 and CH2 ports for (a) TE_0 and (b) TM_0 mode inputs.

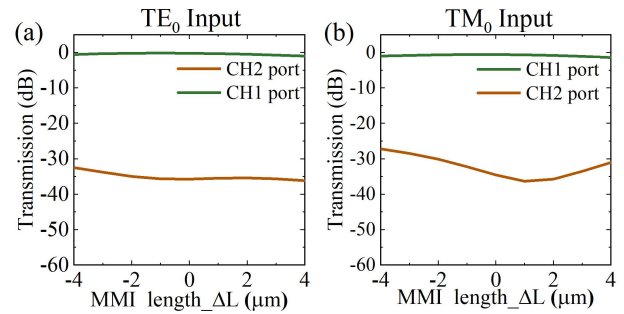


Fig. 5. Transmission spectra at different MMI lengths: (a) TE_0 mode input and (b) TM_0 mode input.

mode conversion process during the entire PRS operation of the device. Therefore, the polarization extinction ratio of the TM_0 mode is more sensitive, but it does not have much impact on the bandwidth of the device.

The designed PRS is fabricated on a commercial X-cut LNOI wafer (NANOLN), which consists of a 400-nm -thick lithium niobate (LN) layer bonded to a $2\text{-}\mu\text{m}$ -thick buried silicon dioxide (SiO_2) layer on a silicon substrate. The LN ridge waveguide structure is defined using standard electron-beam lithography (EBL) on a ZEP520 photoresist, and the pattern is transferred to the LN layer by dry etching. Argon ion (Ar^+) plasma etching is employed to achieve a ridge etch depth of $200\ \text{nm}$. Following the etching process, a $1\text{-}\mu\text{m}$ -thick SiO_2 cladding layer is deposited on the chip surface using plasma-enhanced chemical vapor deposition (PECVD).

Figure 6 shows the scanning electron microscopy (SEM) image and a schematic of the PRS test structure. The fabricated PRS comprises two adiabatic tapers, a straight waveguide, two s bends, a phase shifter, and a 2×2 MMI coupler, as displayed in Fig. 6(a). Figure 6(b) displays the cross-sectional SEM image of the input waveguide, revealing a smooth sidewall, which indicates high fabrication quality. Figure 6(c) illustrates the experimental setup for device characterization. The experimental setup includes a tunable laser source (Keysight 81940A), a polarization controller (PC), a lensed fiber, and an optical power meter (Keysight N7745A). Light from the tunable laser source passes through the PC and is coupled into the PRS via a lensed fiber (LF). The optical power at both output ports is measured using an optical power meter. Since the laser operates within the $1520\text{--}1630\ \text{nm}$ wavelength range, only a $110\ \text{nm}$ segment was tested in the experiment.

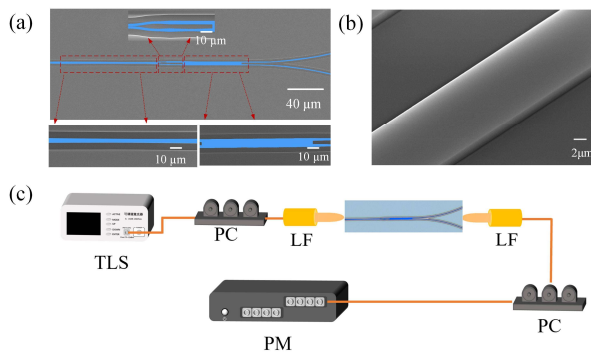


Fig. 6. (a) SEM image of the fabricated PRS. (b) SEM image of the cross section of the input waveguide. (c) Schematic diagram of the device structure used for testing within the 1520–1630 nm wavelength range.

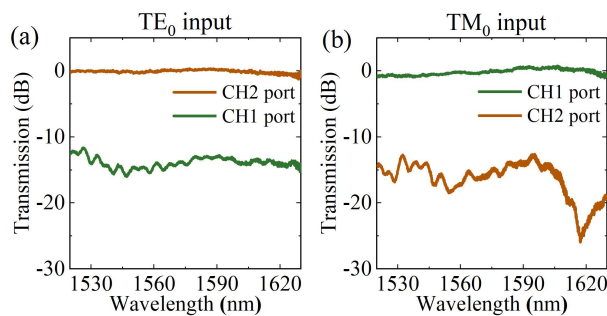


Fig. 7. Experimental transmission at CH1 and CH2 ports for (a) TE_0 and (b) TM_0 mode inputs.

Table 1. Comparison of PRS Performance on TFLN

Ref	Length (μm)	PER (dB)	Max. IL (dB)	Fabrication Tolerance (nm)	Bandwidth (nm)
[13]	8000	~8TE ~10TM	2.0TE 3.0TM	N.A.	130 (1500–1630)
[14]	>1200	>10.0TE >10.0TM	1.5TE 1.0TM	150	50 (1520–1570)
[15]	440	>26.6TE >19.6TM	1.0TE 1.4TM	N.A.	60 (1520–1580)
This work	228	>12.6TE >12.6TM	1.0TE 1.0TM	400	110 (1520–1630)

As shown in Fig. 7, the insertion loss of the PRS at 1550 nm is approximately 0.2 dB for the TE_0 mode input and 0.6 dB for the TM_0 mode input. In the 1520–1630 nm wavelength range, the measured insertion losses for both the TE_0 and TM_0 modes are less than 1.0 dB, while the PER for both modes exceed 13 dB. This performance spans the telecommunication C- + L-band and aligns with the theoretical results. The fabricated devices exhibit a maximum insertion loss of only 0.3 dB in the TE_0 mode within the 1520–1600 nm range. This demon-

strates the high efficiency and performance of the PRS, making it suitable for high-capacity optical communication systems, including those that utilize polarization-division multiplexing (PDM).

Table 1 summarizes recent advancements in thin-film lithium niobate polarization rotator-splitters (PRSs). Among the devices listed, the PRS proposed in this paper exhibits the shortest length, with insertion loss and bandwidth comparable to those of other studies. Additionally, the proposed PRS achieves high performance while minimizing device size, maintaining an excellent balance between bandwidth and insertion loss.

We experimentally demonstrated a wideband and compact PRS on the TFLN platform. The device consists of two adiabatic tapers, a phase shifter and a 2×2 MMI coupler with general interference. The fabricated PRS was compact, with a length of 228 μm. Experimental measurements indicate that the fabricated device achieves a bandwidth of 110 nm, covering both the C-band and L-band. At a wavelength of 1550 nm, the insertion losses are as low as 0.2 dB for the TE_0 mode and 0.6 dB for the TM_0 mode. The polarization extinction ratios (PERs) exceed 13 dB. Additionally, the device exhibits excellent robustness to process tolerances within a ± 4 μm range. This work lays the foundation for polarization-multiplexed transceivers, reconfigurable optical networks, and quantum photonic systems on TFLN platforms.

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Data availability. Data may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

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